NCFE Level 2 Functional Skills Qualification in Mathematics (603/5060/X)

Paper number:	P001258
Section B:	Calculator Test



Assessment window: Monday 14 October 2019 – Friday 18 October 2019
Time allowed: 1 hour 30 minutes

Learner instructions

- Answer **all** questions.
- Read each question carefully.
- Write your answers in the spaces provided.
- Show your working, as marks may be awarded for working.
- State units in your answers, where appropriate.
- Check your work.
- Use $\pi = 3.14$

Learner information

- Section B contains **Activities 2, 3** and **4**.
- The maximum mark for this section is **45**.
- The marks available for **each** question are shown in brackets.

Resources

You will need a:

- pen, with black or blue ink
- pencil and eraser
- 30 cm ruler
- protractor
- calculator.

If extra pages are used, please make sure your name and centre name are on them and they are securely fastened to this booklet.

Please complete the details below clearly and in BLOCK CAPITALS.

Learner name

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- Subject: NCFE Functional Skills Maths Level 2
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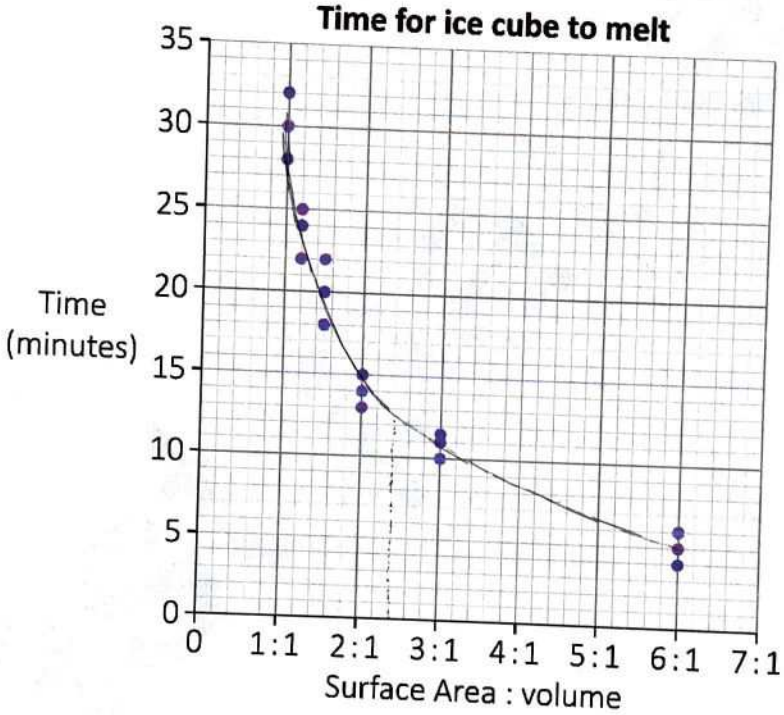
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2 (a) Tom is studying science at college.
He is learning about ice.

Tom does an experiment.
He shows that the ratio of an ice cube's surface area to its volume affects the time it takes for the ice cube to melt.

This scatter diagram shows his results:



Estimate how long this ice cube will take to melt.

[4 marks]

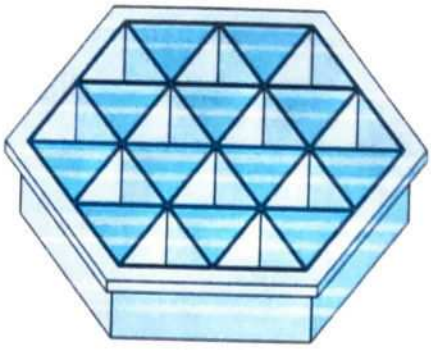
SA: $6 \times 2.5^2 = 37.5\text{cm}^2$.

V: $2.5^3 = 15.625\text{cm}^3$.

$$\frac{37.5}{15.625} = 2.4$$

Your answer: 12.5 minutes

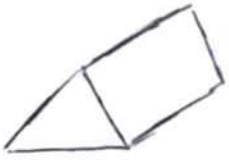
this mould for making 24 triangular p



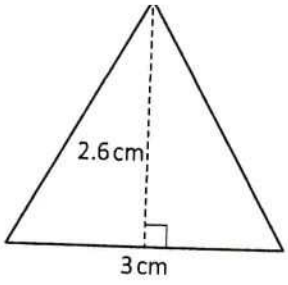
Mould

Draw a triangular prism.

[1 mark]



...prism have base ... cm and height 2.6



Not drawn accurately

Tom fills the hexagonal mould with water to a depth of 3.5 cm

Tom knows that:

- to calculate the volume of each triangular prism, he can use the formula

  Volume = area of triangular face × depth of prism

- when a volume of water freezes, the ice that is formed has a volume 4% more than the volume of the water.

What will be the **total** volume of ice formed when all the water in the hexagonal mould freezes?

[5 marks]

$\frac{1}{2} \times 2.6 \times 3 = 3.9 \text{ cm}^2$.

$3.9 \text{ cm}^2 \times 3.5 \text{ cm} = 13.65 \text{ cm}^3$

$13.65 \times 24 = 327.6 \text{ cm}^3$

$1.04 \times 327.6 = 340.704 \text{ cm}^3$

Your answer: 340.704 **cm³**

Please turn over

e that float in



Tom reads that the icebergs float because the density of ice is less than the density of seawater.

The density of ice is 0.92 g per ml and the density of the sea is 1.03 g per ml

Calculate 0.92 as a percentage of 1.03

Give your answer as a percentage to the nearest whole number.

[2 marks]

$\frac{0.92}{1.03} \times 100 = 89.32\%$

Your answer: 89 %

2 (e) Tom reads that scientists are planning to move an iceberg across the sea to provide water to a very dry country.

The iceberg weighs 40 million tonnes.

Write 40 million using numbers.

[1 mark]

Your answer: 40 000 000

'els.

By the time it arrives, the iceberg will weigh only 25 million tonnes.

By what fraction will the original iceberg have reduced in weight by the time it arrives?

Give your answer in its simplest form.

[2 marks]

$\frac{40-25}{40} = \frac{15}{40} = \frac{3}{8}$.

Your answer: $\frac{3}{8}$

[Total marks: 15]

Past Paper

Please turn over

3 (a) Faisal starts a new job.

He will work 37 hours a week for 48 weeks a year and is paid £8.25 an hour.

He must pay:

- 0% income tax on the first £12 500 he earns each year
- 20% income tax on anything over £12 500 that he earns.

His income tax payments are spread evenly over the 48 weeks.

Faisal's first weekly wage slip shows that he has paid £8.96 income tax.

He thinks he has paid more than he should.

Is he correct?

Show how you decide.

[4 marks]

$37 \times 48 \times £8.25 = £14652$

$£14652 - £12500 = £2152$ taxable.

$0.2 \times £2152 = £430.40$.

$\frac{£430.4}{48} = £8.96\dot{}$

$\Rightarrow £8.97$.

Your answer: No.

he 5 : 3 : 2 rule.

The rule says you should spend your wages after tax in the ratio 5 : 3 : 2 on **needs : wants : savings**

- **needs** are bills like rent that must be paid
- **wants** are extra things like social life and new clothes which are not needed
- **savings** are amounts of money kept for the future.

Sonya earns £480 a week after tax.

Last week she spent £185 on going out with friends.

Was this in line with the 5 : 3 : 2 rule?

Show how you decide.

[2 marks]

$\frac{3}{5+3+2} = \frac{3}{10}$.

$\frac{£185}{£480} = 0.385. > \frac{3}{10}$

Your answer: No.

ach month for his mortgage and othe

This is 45% of his monthly income.

What is Dec's monthly income?

Give your answer to the **nearest ten pounds.**

[2 marks]

$\frac{£908.64}{0.45} = £2019.20$

$\rightarrow £2020$

Your answer: £ 2020

ness claims that the mean salary in t
the last year.

One of the workers says that the gap between the highest and lowest earners has more than doubled.

Use this data about the company salaries to decide whether **each** of the statements is true. Show your working.

[5 marks]

Annual salaries (£s)							
Last year (12 employees)				This year (14 employees)			
15 275	42 122	15 679		15 500	33 775	17 995	34 065
20 000	25 344	25 879		21 339	20 112	24 995	23 115
85 682	15 998	53 245		157 250	40 631	45 888	
55 333	70 507	32 196		60 200	80 555	17 160	

15275+42122+...+70507+32196 = 457260

$\frac{457260}{12} = 38105$

15500 + 33775 ... + 80555 + 17160 = 592580

$\frac{592580}{14} = 42327.14.$

$100 \times \frac{42327.14 - 38105}{38105} = 11.08\%$ increase.

Range (LY):
85682 − 15275
= 70407

Range (TY):
157250 − 15500 =
141750.

70407 × 2 = 140814.

Your answer: Both statements are correct.

Please turn over

Exal

He is paid a basic salary plus a bonus based on the sales he makes.

Last year Dave's bonus was $\frac{3}{8}$ of his salary.

Write $\frac{3}{8}$ as a decimal **and** as a percentage.

[2 marks]

Decimal:

0·375

Percentage:

37·5%.

[Total marks: 15]

Past paper

se only

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Please turn over for the next activity.

Past Paper

Exam

4 (a) Archaeologists study human history by digging up and analysing items from the past.

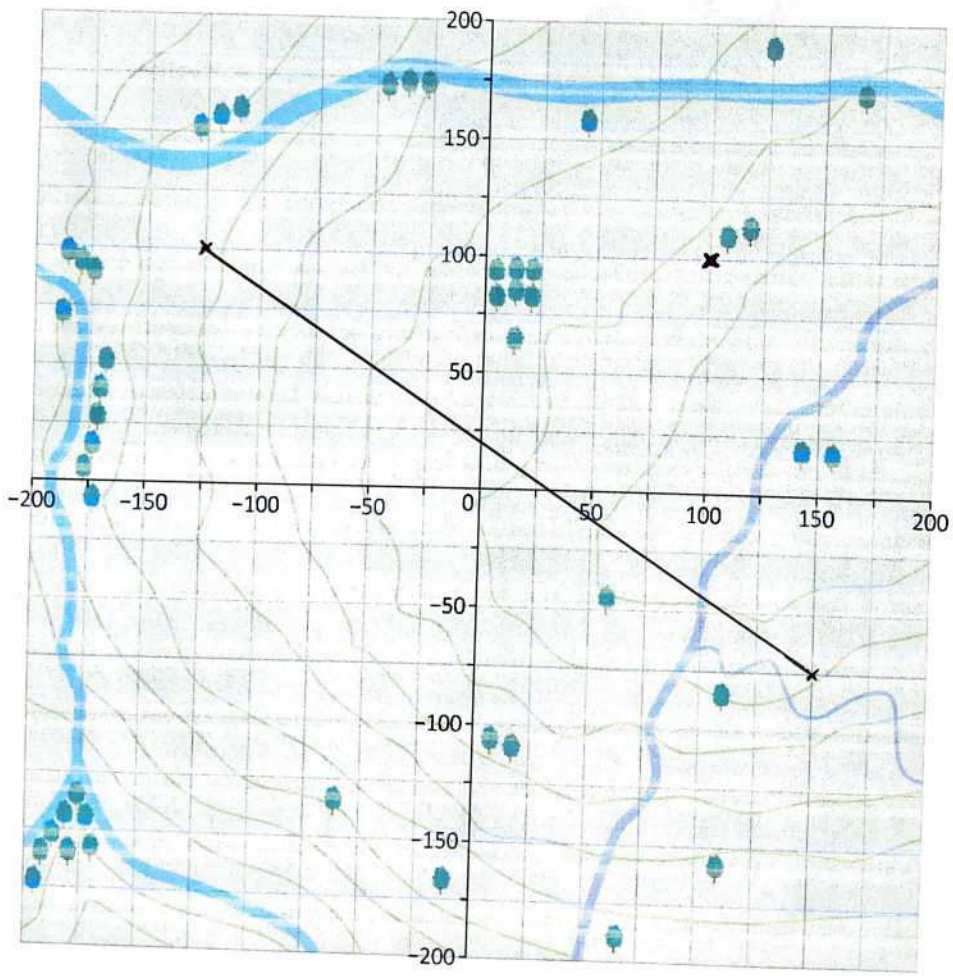
Chen is a volunteer with a community archaeology group.
At the weekends she helps to dig at an archaeological site.

The volunteers have to record exactly where each item is found.
They do this by using a coordinate grid laid over a map of the site.

The volunteers have found the wall of a building.
It runs in a straight line from (–125, 100) to (150, –75)

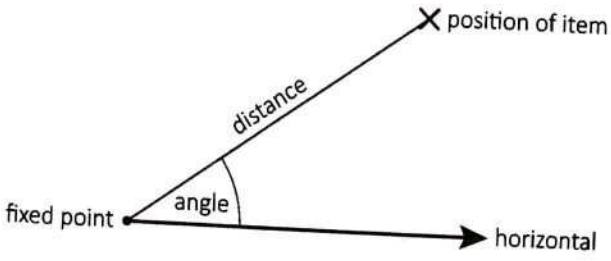
Draw the position of the wall on the grid.

[2 marks]



volunteers can record the position of a

- measure the object's distance from a fixed point and
- measure the angle (see example in the diagram below).



The volunteers find an object at the point (100, 100) on the coordinate grid.

i) Mark the point on the grid in **4 (a)**.

The volunteers measure the distance of the object from the fixed point (0, 0)
They then measure the angle.

ii) What size is the angle?

[2 marks]

$\frac{100-0}{100-0} = 1.$ $\tan^{-1} 1 = 45°.$

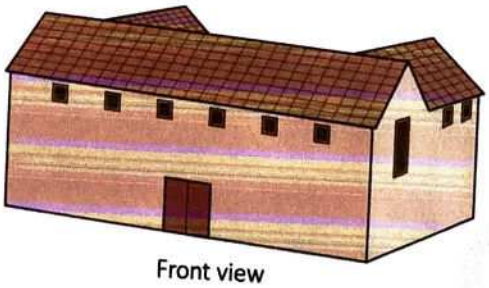
Your answer: 45 **degrees**

Please turn over

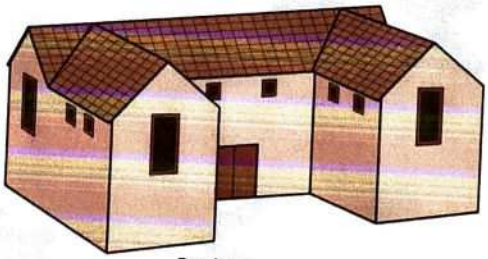
ı building.

(d)

They think the building used to look like this:



Front view

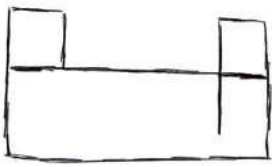


Back view

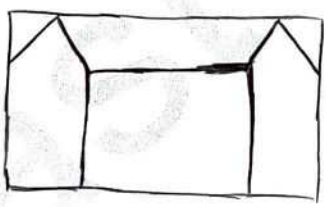
- the side elevation of the building.

[2 marks]

Plan view:



Side elevation:



Past paper

otprints that were preserved in volcar

Ex

The foot length was 18.8 cm and the stride length (the distance between the footprints) was 164.5 cm

Archaeologists use the formula: $R = \frac{\text{stride length}}{4 \times \text{foot length}}$

Chen uses a calculator to calculate R for the footprints found in the volcanic ash.

She enters 164.5 ÷ 4 × 18.8 and gets the answer 773.15

Explain what Chen has done wrong.

[1 mark]

BIDMAS dictates that $\frac{164.5}{4}$ was done first, then multiplied by 18.8.

She should enter "164.5 ÷ (4×18.8)".

son was probably walking.
person was probably running.
If R is in between 2.0 and 2.9 it is not possible to decide.

Tick the correct statement about the person whose footprints were preserved in the volcanic ash.

- [ ] They were probably walking.
- [ ] They were probably running.
- [x] It is not possible to decide.

Explain your choice.

[2 marks]

$$\frac{164.5}{4 \times 18.8} = 2.1875$$

rse only

community

E)

The probability that they all attend on a Saturday is 0.35
The probability that they all attend on a Sunday is 0.12

The site is open next weekend.

Calculate the probability that all the volunteers will attend on both Saturday and Sunday next weekend.

Give your answer as a percentage.

[2 marks]

$0.35 \times 0.12 = 0.042$

$= 4.2\%$

Your answer: 4.2 %

'day and Sunday fo' 20 weekends di

This table shows the number of days each volunteer attends out of the 52 days available.

The information for Chen is missing.

Volunteer	Number of days
Ada	48
Barry	25
Tim	30
Chris	24
Gwen	43
Sia	39
Rita	40
Rav	18
Jon	35
Chen	

Chen says that the mode of the 9 values is 48

Is she correct?

Explain your answer.

[1 mark]

There is no mode, each value appears once only.

...r the 10 volu... ...including Chen wa...

What fraction of the days did Chen attend?

Give the fraction in its simplest form.

[3 marks]

5th (in list of 9) is 35.

$$\frac{\text{Chen's value} + 35}{2} = 33.5$$

$$\Rightarrow \text{Chen's value} = 32.$$

$$\frac{32}{52} = \frac{8}{13}$$

Your answer: $\frac{8}{13}$.

[Total marks: 15]

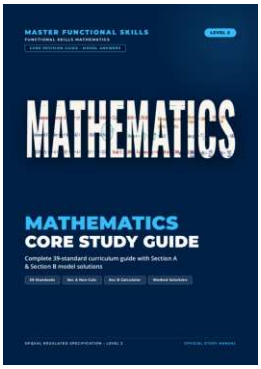
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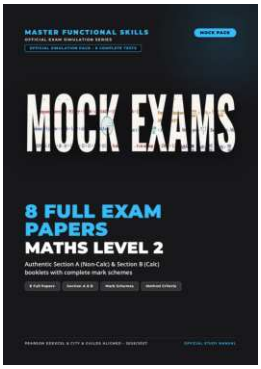


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